Lab 1.2 AI versus engineer challenge

IES | 15 minutes | LO2, LO3

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

Inputs and preparation

Open the mock-data folder: problem-statement.txt, claim-register.csv. Use prompts.md for copy and paste. Save your reviewed work under outputs. Earlier lab outputs are optional; this folder provides standalone inputs.

Procedure

1. Open problem-statement.txt. In pairs, assign one learner to write manually and the other to use Copilot Chat. Start the same five-minute timer and use the same source.

2. Prepare a technical brief with context, observations, unknowns, hypotheses, evidence requests and decision owner. The AI participant uses the prompt below; the manual participant uses the same headings.

3. Compare each factual statement with the source. Mark unsupported assertions in claim-register.csv. In particular, distinguish a reported vibration from a measured vibration value.

Lab 1.2 continued

4. Ask Copilot to repair unsupported claims. Record original draft time, review time and revision time separately. Do not compare draft speed alone.

5. Save the corrected brief and a three-sentence comparison. Identify one decision that remains with an engineer and one repeatable drafting task suited to AI.

Check before moving on

Facts match the notes and unknowns stay unknown.

Timing includes human review and the brief has a decision owner.

Access or source problems

Use approved paired access or the local source files. Record unavailable live capabilities as not executed. If an attachment is not supported, paste the short synthetic text or reference its approved SharePoint copy. Do not use personal accounts to bypass controls.

Lab 1.2 prompts and evidence

Main prompt

Act as an engineering documentation assistant. Convert problem-statement.txt into a 250-word technical brief. Use headings Context, Observations, Hypotheses, Missing evidence and Review decision. Cite the source and preserve timestamps and units. The suggestion to replace the pump is unverified. Do not turn it into an approved action.

Verification prompt

Audit the draft against the named source files. List each claim, source location, uncertainty and correction. Recalculate key values independently. Return a revised draft and change log; do not approve the engineering deliverable.

Expected output example

Illustrative teaching example, not a recorded AI run. Observation: 32 m3/h at 09:12, returning to 47 m3/h at 09:20. Hypothesis: measurement or process disturbance, not a confirmed defect. Request: suction pressure and calibration evidence. Decision: responsible engineer to review.

Retain your evidence

Keep the source versions, prompt, tool, output, corrections, independent checks and reviewer decision. Record any capability or permission test that was not executed. Use templates/prompt-log.csv. Do not publish real workplace data.